



LECTURE #12

Continuing: Processing Raster Data

OBJECTIVE

- CALCULATE THE NDBI (BUILT-UP INDEX) AND APPLY ADVANCED RASTER PROCESSING TECHNIQUES (MASKING AND VECTORISATION) USING SENTINEL IMAGES.

STEPS

1. UNDERSTAND THE NDBI (BUILT-UP INDEX).
2. CALCULATE THE NDBI USING QGIS.
3. CREATE A BUILT-UP MASK.
4. CONVERT RASTER MASK TO VECTOR (RASTER TO POLYGON)
5. IMPROVE RASTER SYMBOLOGY

1 – UNDERSTAND THE NDBI (BUILT-UP INDEX)

NDBI HIGHLIGHTS SURFACES THAT ARE TYPICALLY IMPERVIOUS, MEANING THEY DO NOT ABSORB WATER AND ARE USUALLY MAN-MADE.

THESE SURFACES (CONCRETE, ASPHALT, ROOFTOPS) TEND TO REFLECT LIGHT DIFFERENTLY FROM NATURAL SURFACES.

BUILT-UP AREAS GENERALLY HAVE:

- HIGHER REFLECTANCE IN THE SWIR BAND
- LOWER REFLECTANCE IN THE NIR BAND

THIS CONTRAST ALLOWS US TO DISTINGUISH URBAN AREAS FROM NATURAL ONES.

NORMALIZED DIFFERENCE BUILT-UP INDEX

IS USED IN REMOTE SENSING TO MEASURE THE IDENTIFY AND MAP BUILT-UP AREAS, SUCH AS BUILDINGS, ROADS, AND OTHER ARTIFICIAL SURFACES..

NDBI COMPARES SWIR AND NIR CONSIDERING:

- BUILT-UP / IMPERVIOUS SURFACES → HIGH SWIR AND LOW NIR
- BARE SOIL OR MIXED AREAS → SWIR SLIGHTLY HIGHER THAN NIR → VALUES NEAR ZERO
- VEGETATION / WATER → HIGH NIR AND LOW SWIR → RESULTS NEGATIVE
-

$$NDBI = \frac{SWIR - NIR}{SWIR + NIR}$$

NDBI WORKS BECAUSE BUILT-UP SURFACES REFLECT MORE SHORT-WAVE INFRARED (SWIR) LIGHT THAN NEAR-INFRARED (NIR).

BUILT-UP MATERIALS HAVE PHYSICAL AND CHEMICAL PROPERTIES THAT INCREASE SWIR REFLECTANCE, WHILE THEIR NIR REFLECTANCE REMAINS RELATIVELY LOW. THE STRONGER THIS DIFFERENCE, THE HIGHER THE NDBI VALUE, INDICATING A GREATER PRESENCE OF IMPERVIOUS, URBANIZED SURFACES.

1 – UNDERSTAND THE NDBI (BUILT-UP INDEX)

02

$$\text{NDBI} = \frac{\text{SWIR} - \text{NIR}}{\text{SWIR} + \text{NIR}}$$

BAND	NAME
Band 11 (20 m) or Band 12 (20 m)	SWIR (Short Infrared)
B08 (10m resolution)	NIR (Near Infrared)

ARE THERE TWO
SWIR BANDS IN
SENTINEL??

YES!!

IN PHYSICS, THE SHORT-WAVE INFRARED
(SWIR) REGION IS DEFINED AS:

SWIR = APPROXIMATELY 1.4 MM TO 3.0 MM

HERE IS THE CONTEXT WITHIN THE INFRARED
SPECTRUM:

Spectral Region	Approximate Range
NIR – Near Infrared	0.7 – 1.4 μm
<u>SWIR – Short-Wave Infrared</u>	<u>1.4 – 3.0 μm</u>
MWIR – Mid-Wave Infrared	3 – 8 μm
LWIR – Long-Wave Infrared	8 – 15 μm

SENTINEL-2 INCLUDES TWO SWIR BANDS:

BAND 11 (SWIR 1) → 1.565 – 1.655 MM

BAND 12 (SWIR 2) → 2.100 – 2.280 MM

BOTH FALL INSIDE THE PHYSICAL SWIR RANGE.

1 – UNDERSTAND THE NDBI (BUILT-UP INDEX)

BOTH B11 AND B12 AREA SWIR BUT THEY ARE NOT IDENTICAL, EACH HIGHLIGHTS DIFFERENT PROPERTIES:

BAND 11 (SWIR 1 – 1.61 MM)

MORE SENSITIVE TO MOISTURE CONTENT IN SOIL AND VEGETATION.

BETTER SEPARATION BETWEEN VEGETATION VS BUILT-UP.

PRODUCES MORE STABLE NDBI RESULTS IN MANY CASES.

BAND 12 (SWIR 2 – 2.19 MM)

MORE SENSITIVE TO MINERALS AND DRY MATERIALS. ENHANCES BARE SOIL AND DRY URBAN SURFACES.

SOMETIMES EXAGGERATES SOIL BRIGHTNESS, GIVING MORE NOISE IN NDBI.

RECOMMENDED FOR MOST NDBI APPLICATIONS:

USE BAND 11 (SWIR 1).

IT GIVES CLEANER DIFFERENTIATION BETWEEN:

- VEGETATION
- BUILT-UP
- WATER

AND IS WIDELY USED IN URBAN REMOTE SENSING LITERATURE (ACADEMIC PAPERS).

USE BAND 12 WHEN:

- THE STUDY AREA HAS VERY DRY SOILS
- YOU WANT TO EMPHASIZE DIFFERENCES BETWEEN CONCRETE, ASPHALT AND BARE SOIL
- YOU ARE COMBINING NDBI WITH MINERAL OR SOIL ANALYSIS

2 – PROCESS NDBI IN QGIS

STEPS:

1. LOAD THE BANDS INTO QGIS;
MENU LAYER → ADD LAYER → ADD RASTER LAYER
2. DEFINE THE DESIRED SPATIAL RESOLUTION
(**NEED TO RESAMPLE** IF YOU WANT 10M). 
3. RUN NDBI FORMULA IN RASTER CALCULATOR.

WHAT IS A RESAMPLING?

RESAMPLING IS THE PROCESS OF CHANGING THE SPATIAL RESOLUTION OR GRID ALIGNMENT OF A RASTER DATASET.

IN SIMPLER TERMS, IT MEANS ADJUSTING THE SIZE OF THE PIXELS SO THAT TWO OR MORE RASTER LAYERS MATCH EACH OTHER.

QGIS CAN INTERNALLY HANDLE MIXED RESOLUTIONS (BUT NOT PERFECTLY), SO, WE SHOULD BETTER RESAMPLE B11 TO 10M RESOLUTION.

RASTER → PROJECTIONS → WARP (REPROJECT)

"OUTPUT FILE RESOLUTION" FIELDS:

- X RESOLUTION: 10
- Y RESOLUTION: 10

METHOD: CUBIC OR BILINEAR

2 – PROCESS NDBI IN QGIS

STEPS:

1. LOAD THE BANDS INTO QGIS;
MENU LAYER → ADD LAYER → ADD RASTER LAYER
2. DEFINE THE DESIRED SPATIAL RESOLUTION
(**NEED TO RESAMPLE** IF YOU WANT 10M).
3. RUN NDBI FORMULA IN RASTER CALCULATOR. 

MENU RASTER → RASTER CALCULATOR

$$NDBI = \frac{S_{B11} - S_{B8}}{S_{B11} + S_{B8}}$$



("B11_SENTINEL_RESAMPLED@1" - "B8_SENTINEL@1") /
("B11_SENTINEL_RESAMPLED@1" + "B8_SENTINEL@1")

3 – CREATE A BUILT-UP MASK

A BUILT-UP MASK IS A BINARY RASTER THAT SEPARATES URBAN AREAS (1) FROM NON-URBAN AREAS (0).

IT IS CREATED BY APPLYING A THRESHOLD TO THE RASTER.

THE MOST COMMON THRESHOLD IS:

NDBI > 0 → BUILT-UP

NDBI ≤ 0 → NON-BUILT-UP

.
YOU CAN ADJUST THE THRESHOLD DEPENDING ON YOUR CITY, BUT **0 IS THE CLASSICAL STARTING POINT.**

OPEN THE RASTER CALCULATOR
IN QGIS:

RASTER → RASTER CALCULATOR

BUILT-UP MASK USING NDBI ALONE

IN THE EXPRESSION BOX, TYPE:

("NDBI@1" > 0) * 1

EXPLANATION:

"NDBI@1" > 0 PRODUCES TRUE (1) OR FALSE (0)

MULTIPLYING BY 1 CONVERTS TRUE/FALSE INTO NUMERIC VALUES

RESULTING RASTER:

1 = BUILT-UP

0 = NON-BUILT-UP

3 – CREATE A BUILT-UP MASK

A BUILT-UP MASK IS A BINARY RASTER THAT SEPARATES URBAN AREAS (1) FROM NON-URBAN AREAS (0).

IT IS CREATED BY APPLYING A THRESHOLD TO THE RASTER.

THE MOST COMMON THRESHOLD IS:

NDBI > 0 → BUILT-UP

NDBI ≤ 0 → NON-BUILT-UP

.

YOU CAN ADJUST THE THRESHOLD DEPENDING ON YOUR CITY, BUT **0 IS THE CLASSICAL STARTING POINT.**

OPEN THE RASTER CALCULATOR
IN QGIS:

RASTER → RASTER CALCULATOR

BUILT-UP MASK USING NDBI AND NDVI

IN THE EXPRESSION BOX, TYPE:

("NDBI@1" > 0) AND ("NDVI@1" < 0.2)

NDBI ALONE DOES NOT DIRECTLY MAP 'URBAN AREAS'; IT HIGHLIGHTS SURFACES WITH HIGH SWIR REFLECTANCE, WHICH CAN INCLUDE BARE SOIL, DRY FIELDS AND ROCKS, NOT ONLY BUILDINGS.



Quick Review Questions About Spectral Indices

NDVI

- A) WHAT DO THE VALUES OF NDVI REPRESENT? EXPLAIN THE MEANING OF LOW, MEDIUM, AND HIGH NDVI VALUES.
- B) WHY IS THE NIR BAND ESSENTIAL FOR CALCULATING NDVI?
- C) GIVE TWO PRACTICAL APPLICATIONS OF NDVI IN ENVIRONMENTAL STUDIES.

NDWI

- A) WHAT IS THE MAIN PURPOSE OF NDWI?
- B) WHICH SENTINEL-2 BANDS ARE GENERALLY USED TO COMPUTE NDWI, AND WHY?
- C) MENTION ONE SITUATION WHERE NDWI MIGHT PRODUCE MISLEADING RESULTS.
AN EXAMPLE.

NDBI

- A) WHAT DOES NDBI MEASURE, AND WHY IS IT USEFUL IN URBAN ANALYSIS?
- B) EXPLAIN WHY THE SWIR BAND IS USED IN THE NDBI FORMULA.
- C) HOW WOULD YOU INTERPRET A PIXEL WITH A HIGHLY POSITIVE NDBI VALUE?

COMPARISON

- A) COMPARE NDVI AND NDBI: IN WHAT SITUATIONS MIGHT THESE INDICES BE CONFUSED?
- B) WHY IS IT IMPORTANT TO ANALYZE MORE THAN ONE INDEX WHEN STUDYING AN AREA?
- C) HOW CAN NDVI AND NDBI BE USED TOGETHER TO IMPROVE THE IDENTIFICATION OF URBAN VEGETATION?
- D) WHY MIGHT NDVI VALUES DECREASE IN URBAN AREAS, EVEN WHEN SMALL PATCHES OF VEGETATION EXIST?
- E) EXPLAIN HOW THE SPECTRAL BEHAVIOR OF SOIL CAN AFFECT BOTH NDVI AND NDBI CALCULATIONS.
- F) IN WHAT SCENARIO COULD WATER BODIES INFLUENCE THE INTERPRETATION OF NDVI OR NDBI VALUES?
- G) HOW CAN SEASONAL CHANGES (E.G., DRY VS. RAINY SEASON) ALTER THE INTERPRETATION OF THESE INDICES?
- H) WHEN MAPPING LAND USE, WHY IS IT RISKY TO RELY SOLELY ON NDVI OR SOLELY ON NDBI? GIVE



LECTURE #13

Image Classification

OBJECTIVE

- EXPLORATORY CLASSIFICATION USING K-MEANS.

STEPS

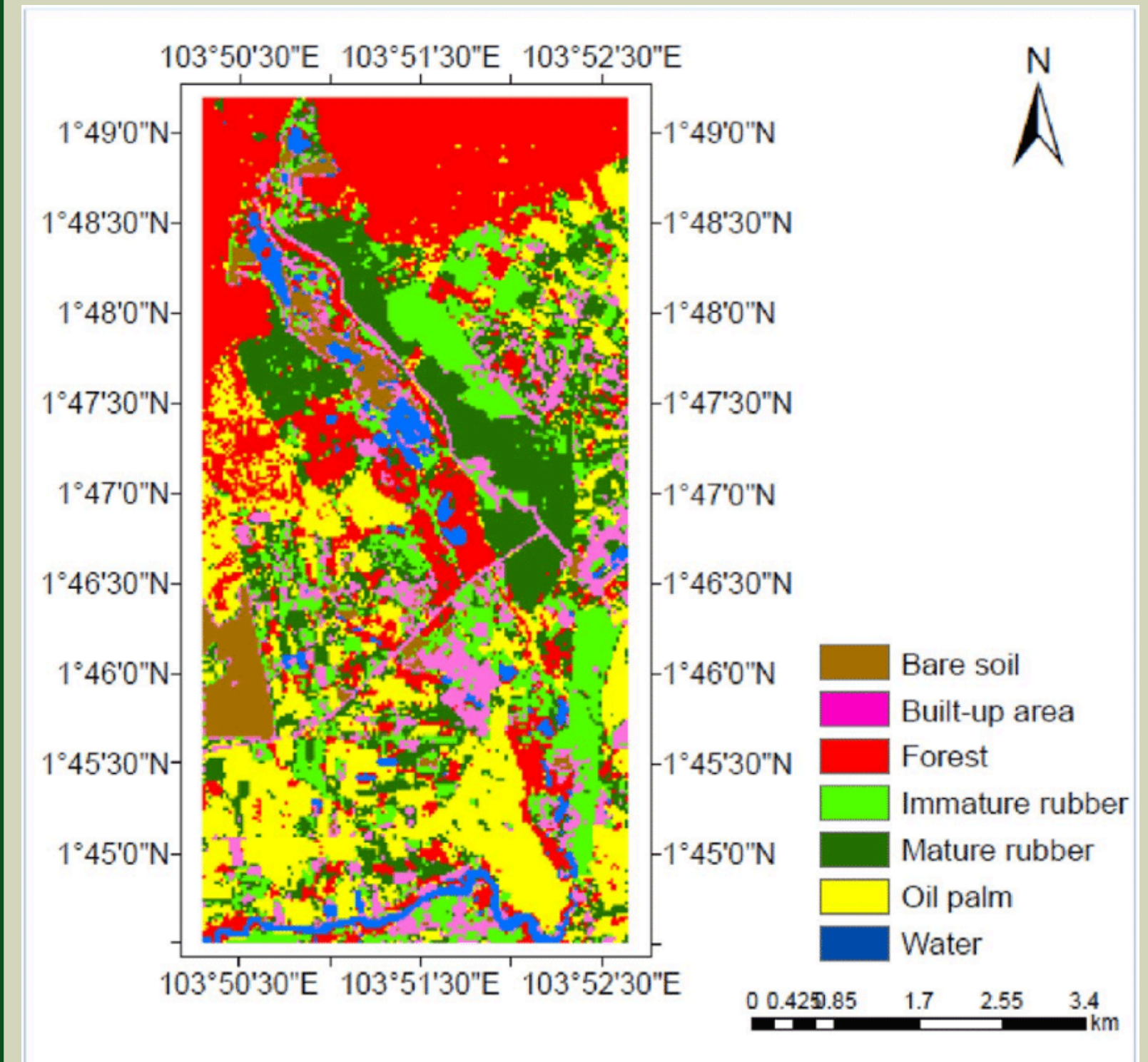
1. WHAT IS IMAGE CLASSIFICATION?
2. TYPES OF IMAGE CLASSIFICATION.
3. 4. UNSUPERVISED CLASSIFICATION WITH SCP
PLUG-IN.

1 – WHAT IS IMAGE CLASSIFICATION?

IMAGE CLASSIFICATION IS THE PROCESS OF **ASSIGNING A THEMATIC LABEL** (E.G., FOREST, URBAN, WATER) TO EVERY PIXEL IN A REMOTELY SENSED IMAGE.

PURPOSE

- LAND-USE/LAND-COVER MAPPING
- MONITORING VEGETATION, WATER, AGRICULTURE
- DETECTING ENVIRONMENTAL CHANGES
- EXTRACTING INFORMATION FROM MULTISPECTRAL DATA



2 – UNSUPERVISED CLASSIFICATION

WHAT IS UNSUPERVISED CLASSIFICATION?

THE ALGORITHM GROUPS PIXELS BASED ONLY ON **SPECTRAL SIMILARITY**, NO USER-DEFINED CLASSES.

WHEN TO USE IT?

- EXPLORATORY ANALYSIS
- WHEN TRAINING DATA IS UNAVAILABLE
- TO DETECT UNKNOWN PATTERNS IN THE IMAGE

3 – UNSUPERVISED CLASSIFICATION IN QGIS

03

YOU'LL NEED TO USE SCP PLUG-IN:

- GO TO MENU PLUGINS → MANAGE AND INSTALL PLUGINS.
 - MAKE SURE SEMI-AUTOMATIC CLASSIFICATION PLUGIN IS INSTALLED AND ACTIVATED.
- A NEW SCP PANEL WILL APPEAR (USUALLY ON THE LEFT SIDE).

Semi-Automatic Classification Plugin



The Semi-Automatic Classification Plugin (SCP) allows for the supervised classification of remote sensing images, providing tools for the download, the preprocessing and postprocessing of images.

Developed by Luca Congedo, the Semi-Automatic Classification Plugin (SCP) allows for the supervised classification of remote sensing images, providing tools for the download, the preprocessing and postprocessing of images. Search and download is available for Landsat, Sentinel-2 images. Several algorithms are available for the land cover classification. This plugin requires the installation of Remotior Sensus, GDAL, OGR, Numpy, SciPy, and Matplotlib. For more information please visit <https://fromgistors.blogspot.com>.

★★★★★ 1185 rating vote(s), 2413903 downloads

3 – UNSUPERVISED CLASSIFICATION IN QGIS

K-MEANS CLUSTERING (UNSURPEIVED CLASSIFICATION)

STEPS IN SCP

1. BUID A BAND STACK

A. LOAD BANDS;

B. BAND RESAMPLE;

C. BAND SET MANAGER (CREATE BAND STACK);

2. RUN K-MEANS:

A. SELECT NUMBER OF CLASSES (E.G., 6)

B. DEFINE ITERATIONS (E.G., 20)

C. RUN K-MEANS

3. VISUALIZE AND INTERPRET CLASSES

STEP 1: BUILD A BAND STACK

A. LOAD SPECTRAL BANDS INTO QGIS

BAND	NAME	RESOLUTION	USE
B2	Blue	10 m	Water detection, atmospheric scattering, complementary to RGB.
B3	Green	10 m	Vegetation contrast, useful for separating soil/urban/vegetation.
B4	Red	10 m	Chlorophyll absorption, helps vegetation vs soil/urban separation.
B5	Red Edge 1	20 m	Detects early vegetation stress; sensitive to chlorophyll variations.
B6	Red Edge 2	20 m	Differentiates vegetation types and crop stages.
B7	Red Edge 3	20 m	Separates dense vs sparse vegetation; useful for environmental mapping.
B8	NIR	10 m	Strong vegetation signal; ideal for biomass and NDVI-like responses.
B8A	Narrow NIR / Red Edge	20 m	Transition band; improves supervised classification accuracy.
B11	SWIR 1	20 m	Moisture content, burned areas, soil/vegetation discrimination.
B12	SWIR 2	20 m	Dry vegetation, bare soil, geology; very good for land-cover classes.

STEP 1: BUILD A BAND STACK

B. RESAMPLE BANDS TO 10M

MENU → RASTER → PROJECTIONS → WARP (REPROJECT)

Raster Projections - Warp (Reproject)

ParametersLog

Input layer

INPUT RASTER

Source CRS [optional]

Target CRS [optional]

Resampling method to use

Nearest Neighbour

Nodata value for output bands [optional]

Not set

Output file resolution in target georeferenced units [optional]

OUTPUT RESOLUTION → 10

Advanced Parameters

Reprojected

OUTPUT RASTER

☒ Open output file after running algorithm

0%

Cancel

AdvancedRun as Batch Process...

RunCloseHelp

STEP 1: BUILD A BAND STACK

C. BAND SET MANAGER (CREATE BAND STACK)

02

MENU → SCP → BAND SET

Band set definitionActive band set 1DateNaTRoot directory

	Band name	Center wavelength	Multiplicative Factor	Additive Factor	Wavelength unit	
1	B2_10M	490.0	1.0	0.0	nm (1 E-9m)	C
2	B3_10M	560.0	1.0	0.0	nm (1 E-9m)	C
3	B4_10M	665.0	1.0	0.0	nm (1 E-9m)	C
4	B5_10M_resampled	705.0	1.0	0.0	nm (1 E-9m)	C
5	B6_10M_resampled	740.0	1.0	0.0	nm (1 E-9m)	C
6	B7_10M_resampled	783.0	1.0	0.0	nm (1 E-9m)	C
7	B8_10M	842.0	1.0	0.0	nm (1 E-9m)	C
8	B8A_10M_resampled	865.0	1.0	0.0	nm (1 E-9m)	C
9	B11_10M_resampled	1610.0	1.0	0.0	nm (1 E-9m)	C
10	B12_10M_resampled	2190.0	1.0	0.0	nm (1 E-9m)	C

DOUBLE-CLICK EACH CELL TO EDIT.

WAVELENGTH UNIT → NM (1E-9 M)

Band quick settings

WavelengthBand orderWavelength unitnm (1 E-9m)Date2020-01-01

Band set tools

☒ Create virtual raster of band set☐ Create raster of band set (stack bands)☐ Build band overviews☐ Band calc expressions

RUN

CREATE VIRTUAL RASTER

STEP 1: BUILD A BAND STACK

C. BAND SET MANAGER (CREATE BAND STACK)

MENU → SCP → BAND SET

Band set definitionActive band set 1Date NaTRoot directory

	Band name	Center wavelength	Multiplicative Factor	Additive Factor	Wavelength unit	
1	B2_10M	490.0	1.0	0.0	nm (1 E-9m)	C: +
2	B3_10M	560.0	1.0	0.0	nm (1 E-9m)	C:
3	B4_10M	665.0	1.0	0.0	nm (1 E-9m)	C:
4	B5_10M_resampled	705.0	1.0	0.0	nm (1 E-9m)	C:
5	B6_10M_resampled	740.0	1.0	0.0	nm (1 E-9m)	C:
6	B7_10M_resampled	783.0	1.0	0.0	nm (1 E-9m)	C:
7	B8_10M	842.0	1.0	0.0	nm (1 E-9m)	C:
8	B8A_10M_resampled	865.0	1.0	0.0	nm (1 E-9m)	C:
9	B11_10M_resampled	1610.0	1.0	0.0	nm (1 E-9m)	C: ↑
10	B12_10M_resampled	2190.0	1.0	0.0	nm (1 E-9m)	C: ↓

DOUBLE-CLICK EACH CELL TO EDIT.

WAVELENGTH UNIT → NM (1E-9 M)

Band quick settingsWavelengthBand orderWavelength unit nm (1 E-9m)Date 2020-01-01

Band set tools☒ Create virtual raster of band set☐ Create raster of band set (stack bands)☐ Build band overviews☐ Band calc expressions

RUN

CREATE VIRTUAL RASTER

SAVE BAND SET



AFTER CREATING THE NEW BAND SET, CLICK HERE TO VISUALIZE

Band	Name	<u>Wavelength (nm)</u>
B2	Blue	<u>490</u>
B3	Green	<u>560</u>
B4	Red	<u>665</u>
B5	Red-edge 1	<u>705</u>
B6	Red-edge 2	<u>740</u>
B7	Red-edge 3	<u>783</u>
B8	NIR	<u>842</u>
B8A	Narrow NIR	<u>865</u>
B11	SWIR-1	<u>1610</u>
B12	SWIR-2	<u>2190</u>

STEP 2: RUN K-MEANS

SCP → BAND PROCESSING → CLUSTERING OF BAND SET

Clustering of band set

Select input band set1

MethodK-means

☒ Distance threshold0.0001000

Number of classes4

Max number of iterations30

NUMBER OF CLASSES: FROM 4 TO 7 (OPTIONAL)

☐ Use value as NoData0

Seed signatures

☐ Seed signatures from band values

☐ Use Signature list as seed signatures

☒ Use random seed signatures

Distance algorithm

☒ Minimum Distance

☐ Spectral Angle Mapping

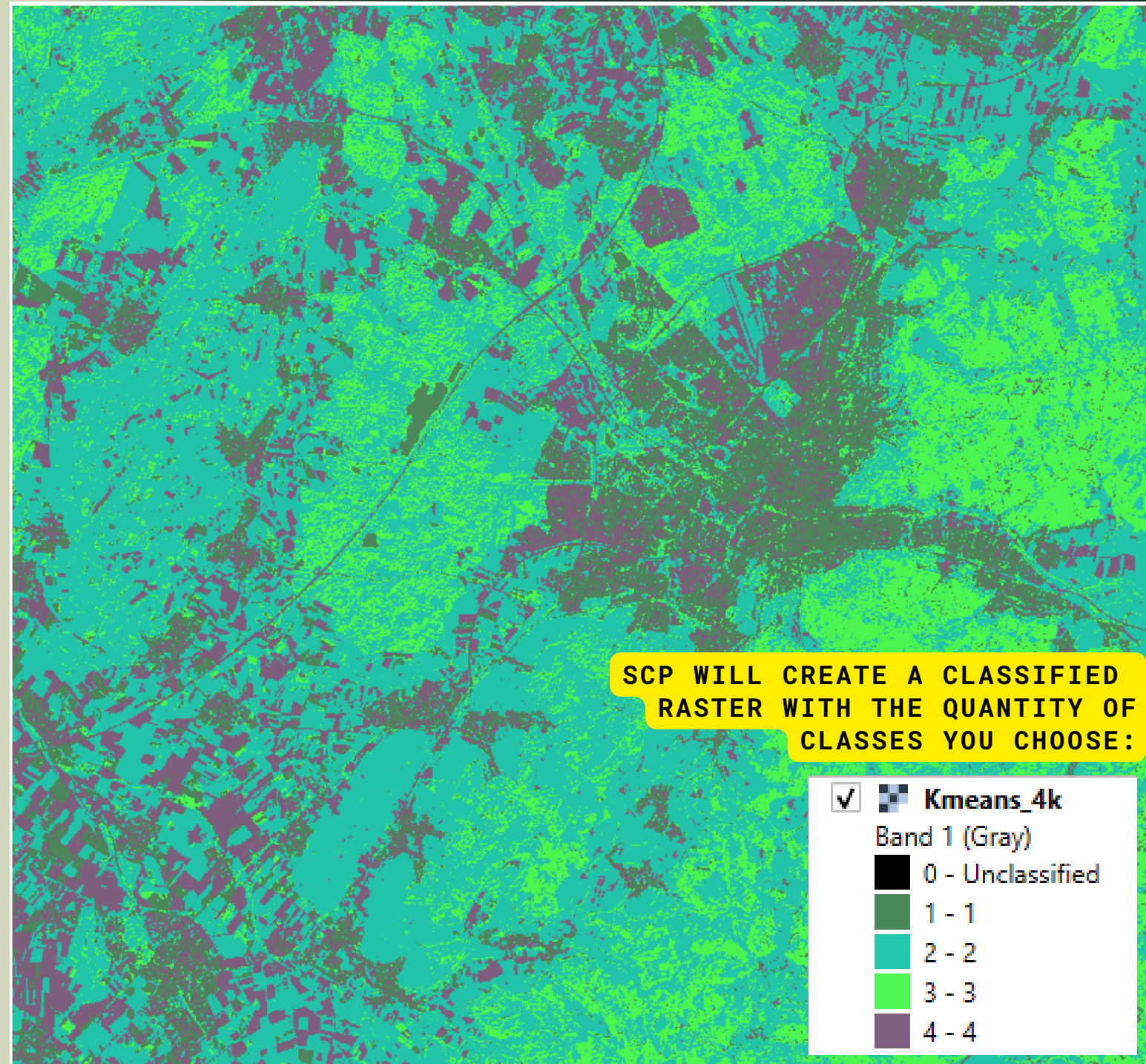
☐ Save resulting signatures to Signature list



CLICK RUN

STEP 3. VISUALIZE AND INTERPRET CLASSES

02



YOU CAN IMPROVE IT...

- TEST K-MEANS WITH DIFFERENT NUMBER OF CLASSES.
- APPLY PROPER SYMBOLOGY
 - RIGHT-CLICK THE OUTPUT RASTER
→ PROPERTIES → SYMBOLOGY